

Tutorial 8: LKM*Nanyang Technological University*

- Q1** What is the purpose of module parameters in kernel modules? How can you define and use module parameters in your loadable module?
- Q2** Write a LKM that allows you to set an integer parameter to control the number of times a message is printed to the kernel log.
- Q3** Discuss the steps and operations involved to allow a user space program, userT8.c to access a LKM. Assume that the user space program is to write a message to a character device, devT8 located in the /dev directory that is registered in the Linux kernel with static major number 103, through a loadable kernel module (LKMT8.ko).
- Discuss the essential program statements in the LKM that are needed to support operation described.
 - Show the relevant commands needed when launching the module and the program.
 - Show the relevant program statements required in the user program userT8.c

that access the I/O device to write a “sc3103T8” message using a buffer defined as follows:
char *msg_u = “sc3103T8”;

- Q4** The following shows the typical boot-up sequence used for the Atmel AT91RM9200 embedded processor. Briefly describe the main functions performed in each of the steps shown.



Figure 8.1: Q2